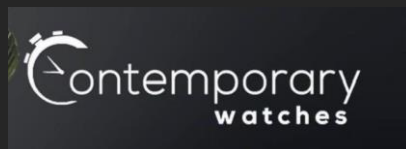


Falcon Force

Repton Dubai F1 Team



Engineering Portfolio



Research and Development



Aerodynamics

This is the study of the way solid objects move through air and how they behave. By studying this, we can understand which shape to make our car in order to make it the fastest we can.

We started our research by looking at the teardrop shape, which is known to be the most aerodynamic shape in the world.

We have used this shape on our wings in our initial design and decided it was very effective and kept it to use in our final car design as well.

Softwares

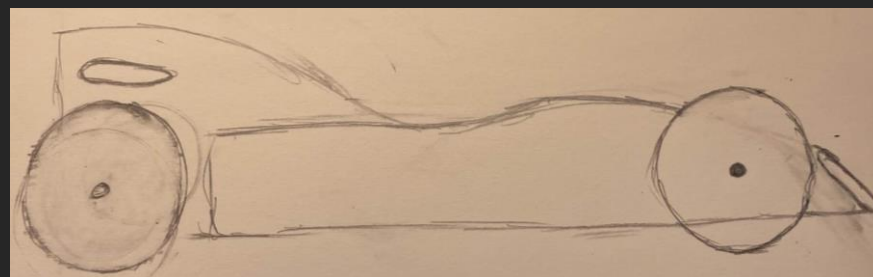
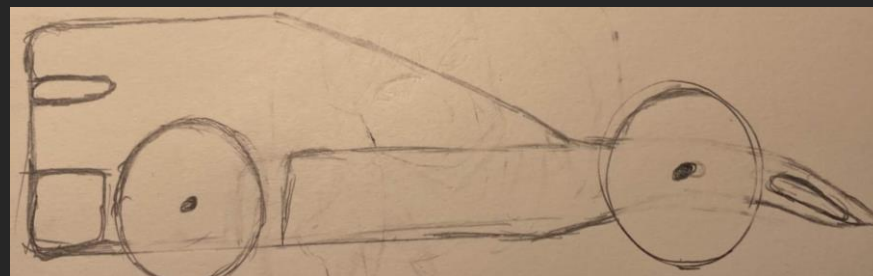
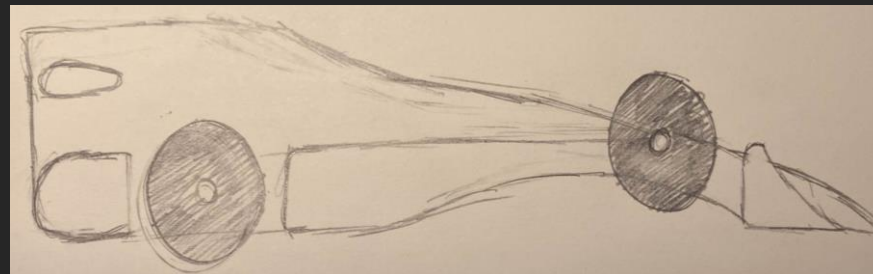
Our team has used CAD to design our car accurately and quickly.

We have designed the initial car using Fusion 360 and later on realised that using solid works will make the designing process easier and quicker for us.



Initial Sketches

Sai and Zac sketched out some different types of sketches for some potential car ideas and ended up creating a car with a few features from each of the sketches shown here.



Weight

As we all know, the lighter an object is, the faster it can move and this is easily provable using the concept of momentum. The lighter the car, the faster it can go.

We've made sure to design our car so that straight after CNC printing, the car weights less than the minimum requirement and later on, we can add layers of paint to make the car exactly 60g as per the technical regulations.



How did we convince sponsors?

Aerodynamics was a very important factor for us to consider throughout each and every step of making and designing the car. Studying aerodynamics enabled us to understand that even one small change in the car could make it extremely fast if done in the right place. Sai and Zac then worked together to analyse car body shapes and wing structure in order to implement it our car designs.

Design Process

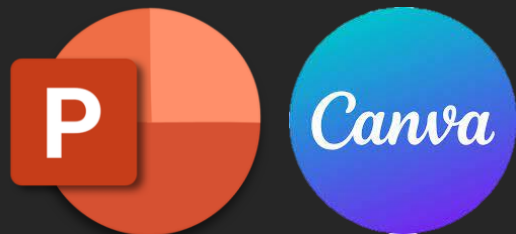


CAD

Our team has used Fusion 360 to design our car as it has the most friendly user-interface and it would be professional as well as easy to understand.

Sai has designed various different cars of which started off with a very different idea and as we tested the car and checked which parts of it could've been improved we ended up with designs looking similar to our final car. Our final car was the car we had made all the modifications to- in order to make the car as streamlined as possible.

For other tasks such as the portfolios, we have used PowerPoint in order to make it easy and understandable for the whole team to use. We all believed that we all had enough experience with PowerPoint and that we can design with ease on this software. Along with PowerPoint, Canva was used to create quick minimalistic posts for our social media.



CAM

Since we are fortunate enough to have a CNC machine in our school- Repton Dubai, we were able to print right here without the hassle meaning we also were able to print multiple versions of the car testing for each one to see which one is faster.



Finishing

Once our team has tested which car we are using, we have sanded the car and brushed multiple layers of paint onto the car to achieve our desired colour and to maintain 60g of weight as per the technical regulations.





The Car Design and Evaluation

Evaluation

We've constantly tested and adapted all of our car designs on a daily basis. We've designed multiple cars and we made sure to check each part of each car. We found that some of our designs were too weak or the front wing was too thin and flimsy. Using information from each test, we changed the designs and finally ended up with the final car we have. In addition to this, our school has the privilege of having an F1 track meaning we could race each car.

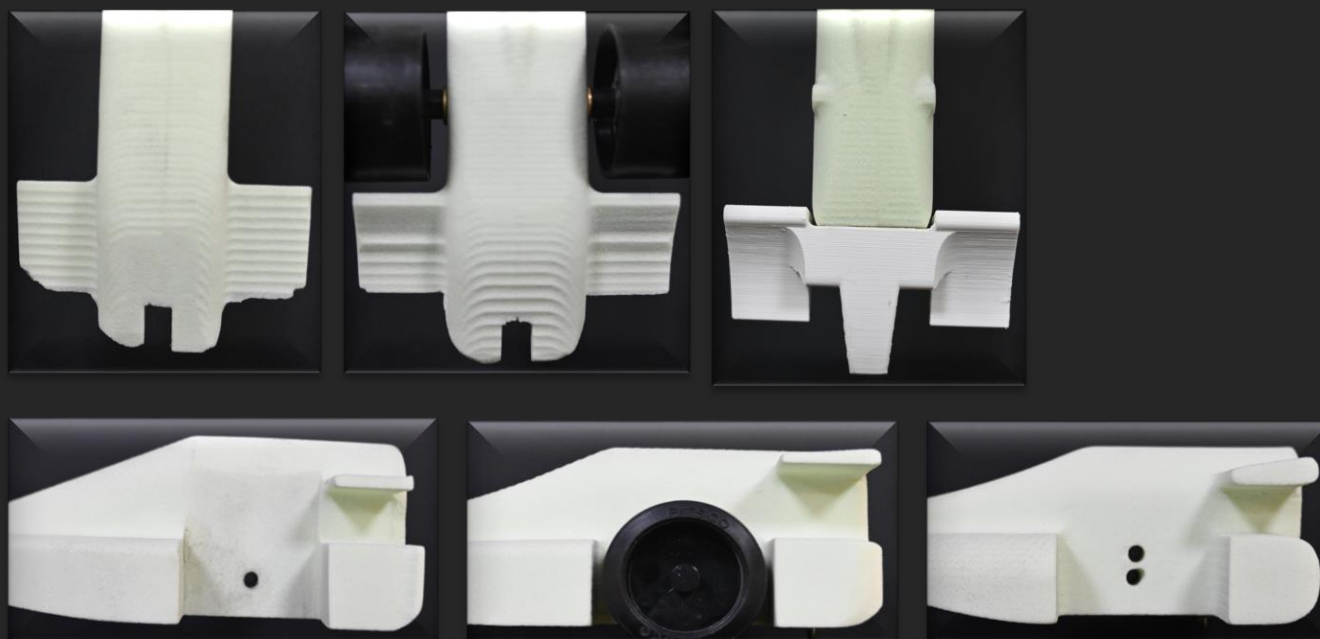
Air Trace

After putting on the layers of paint, we tested our car in the Air Trace to see how aerodynamic it is. We can see from the picture that the fog is moving the way we want it to- above and over the car.

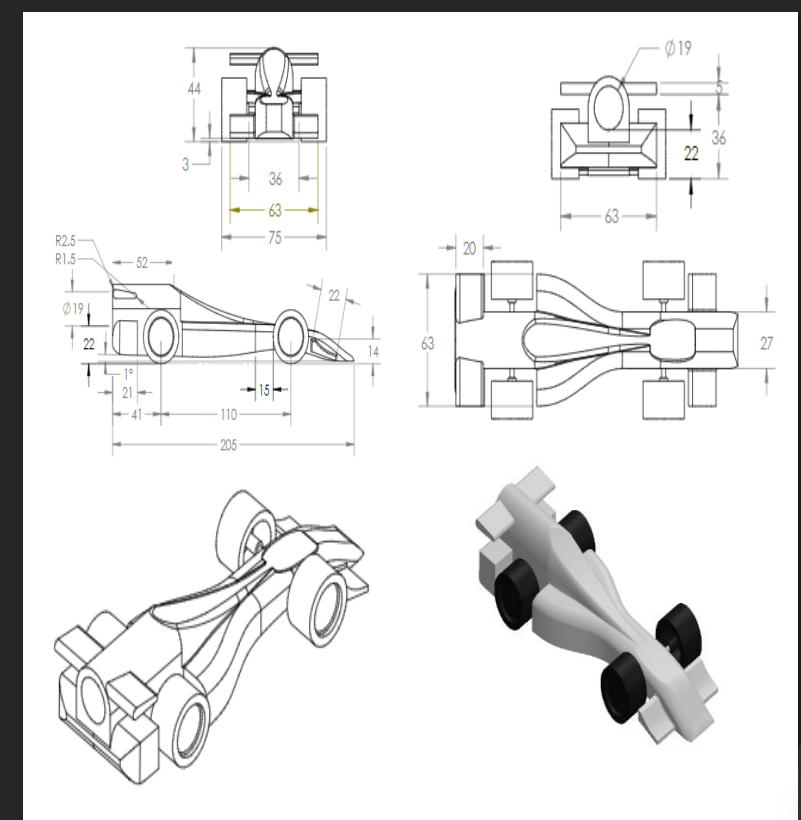


Front and Rear Wing Adaptation

We designed multiple cars and we also printed each car out to physically be able to see any flaws in the designs. We tried multiple versions of front and rear wing adaptations and also tried 3D printing our nose cone.



Orthographic Drawings of Final Car



Challenges and Learnings



Life Skills

It's safe to say that all 6 of our team members have been through a journey full of crests and troughs. We've all learned how to co-operate with each other and interact with people we've never met before. It boosted our communication skills as well as taught us how to deal with communication errors and arguments.

There have been times where the discussion got a little heated up but we've always made sure that whoever is not involved is resolving the issue between the two people. At the end of the day, We are team Falcon Force and we help each other out in the time of need. In order to prevent future fall outs we made a list of all the tasks that had to be done so that nobody had any sort of confusion as to what they had to do and when it was due by. This reduced the rate of arguments by a significant amount and we were able to use time productively rather than wastefully.

A few of the main things we've learnt through this competition are things like how to reach out to sponsors in a professional manner. Furthermore, we had to be able to arrange meetings with various teachers to do various different tasks or to use certain rooms/ equipment such as our school's Nikon suite in which we took all our pictures in. We have found out that splitting up the work evenly and then coming back together is the best strategy for us to work smartly and efficiently.

Software Experience

Our school provides multiple modules throughout the school years where students learn CAD skills. This catered our need for having a good grip on CAD to do this competition. We couldn't have done it without having an ease with the devices and software's we were using.

During the recent Covid-19 pandemic, we all had a forced shift towards technology. This came to our advantage as we now have a lot better understanding of what we're doing rather than just following a set of instructions.



Final Car



Streamline

The most basic yet extremely crucial thing that comes to anybody's mind whilst talking about F1 cars is the car's streamline body. Our team has ensured that we fine tune both ends of our car to make sure it is avoiding as much air resistance as possible.

Rear Wings

One of the most important aspects of an F1 car are its 4 wings. Smallest details matter. This is why Sai and Zac constantly tested the car to make sure the wings were as aerodynamic as possible with minimal amount of weight added.

Branding

Team Falcon Force's logo has been placed so that it is clearly visible on the car to emphasise on our own brand and to create a name in the crowd for ourselves as an F1 in Schools team.

Front Wing

The whole team agreed that the front wing was the most important part of the car as it is one of the first spots on the car that meets air. It was vital that this lifts up the air away from the wheels to avoid drag.

Sponsors

In order to ensure our sponsors are all well advertised, we've placed all of our sponsors in visible locations on the car whilst ensuring we publicise our own brand as well.

